

Physics Extended Essay — Detailed Structure Plan (v2)

Investigation on the frequencies of oscillations of a coupled magnetic–mechanical oscillator

Working document: Edwin’s skeleton + INTERNATIONAL’s evaluation loop • v2, revised and expanded

September 8, 2026 — all placeholder numbers to be regenerated with your own inputs

What is new or changed in v2 (compared with the previous plan)

1. Word budget rebalanced to ≈ 3950 words (the old plan summed to ≈ 4250 — over the limit). Section 5–6 and 10–13 were trimmed; never shrink §10–13 at the expense of anything else.
2. d is now defined unambiguously: record the face-to-face gap g (calipers) and the magnet thickness t , and use the centre-to-centre separation $d = g + t$ everywhere. State this once in Variables and never mix the two.
3. Per-spring *dynamic* calibration promoted to the primary way of fixing k/m : pluck each bare spring and use $f = \frac{1}{2\pi}\sqrt{k/m}$. This removes $E(T)$, the Rayleigh factor 33/140 and the lumped-mass approximation from the theory of f_1 ; the static k becomes a cross-check. It also sharpens the detuning analysis of §10.4.
4. FFT section: add an explicit *detrend* step (subtract the mean / remove the DC peak — the equilibrium offset of the magnets puts a peak at 0 Hz that leaks into the low-frequency region), and add *horizontal* error bars $\Delta(\ln d) = \Delta d/d$, which are comparable to the vertical ones at small d .
5. Magnet tilt / non-coaxiality listed explicitly as a quantified sub-effect of the nonlinear force (one-line estimate + RK4 check), instead of being implicit.
6. Assessment-criteria map added (§8): every section is now tied to a criterion A–E.
7. Suggested writing order added at the end of §6 (write the middle first, intro/conclusion last).
8. Small additions: report reduced χ^2 for each model fit; archive raw videos (not only Tracker exports) as authenticity evidence.

Storyline (keep this on your wall)

Two identical copper leaf springs with repelling magnets form two coupled oscillators. The in-phase mode (f_1) should not feel the magnets; the anti-phase mode (f_2) is stiffened by the magnetic coupling γ . Modelling the magnets as point dipoles gives

$$f_2^2 - f_1^2 = \frac{3\mu_0\mu_m^2}{\pi^3 m d^5},$$

i.e. a d^{-5} power law. f_1, f_2 are measured by video tracking + FFT for $d = 21$ –60 mm. The $\ln$ – $\ln$ gradient was -3.75 ± 0.19 , f_1 drifted by 2.6%, and points beyond 35 mm could not be resolved. The evaluation identifies four systematic effects (FFT resolution/leakage, amplitude-dependent non-linearity of the r^{-4} force, finite magnet size, non-identical springs). A modified method (normal-coordinate tracking, longer records, small release, time-domain two-tone fitting, non-linear least squares) plus an RK4 simulation of the full non-linear equations quantifies each effect and recovers the dipole law within its range of validity.

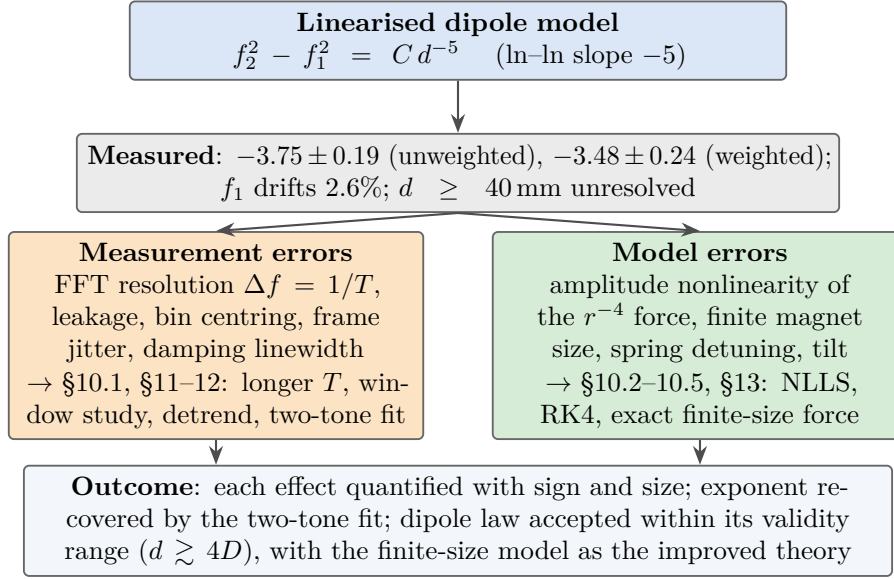


Figure 1: The logic of the essay as a flow chart: one prediction, one measurement, two families of error, one resolved conclusion.

1 Research question

RQ. *How does the equilibrium centre-to-centre separation d of the magnets in a coupled magnetic–mechanical oscillator affect its two normal-mode frequencies f_1 and f_2 , and does the coupling term $f_2^2 - f_1^2$ follow the inverse-fifth-power law predicted by the magnetic-dipole model?*

Why this wording: it names the independent variable (d), the dependent variables (f_1, f_2), the derived quantity that is actually tested ($f_2^2 - f_1^2$) and the theoretical claim (d^{-5}). Keep the title as it is; put the RQ in §1.3.

Definition of d : record the face-to-face gap g with calipers and the magnet thickness t ; the quantity used in every plot and formula is the centre-to-centre separation $d = g + t$. Decide this once, state it in Variables, and never mix the two conventions.

2 Mapping: Edwin’s structure → yours → INTERNATIONAL additions

Word budget is for a 4000-word essay (equations, tables, captions, appendices and references do not count). **Budget rebalanced to ≈ 3950 words to leave a safety margin.** If you overrun, shrink §6.1 and §9 first; *never* shrink §10–13.

	Edwin	Yours (section title)	Words
1 Introduction	§1	1.1–1.3: system (done), modes & beats, motivation & RQ	350
2 Qualitative prediction	§2	§2	70
3 Hypothesis	§3	§3 (quantitative: slope -5)	50
4 Variable table	§4	§4 (table; FFT settings as controlled)	60
9 Theoretical model	§9	§5 (moved before methodology, as INTERNATIONAL does; +§5.4 remarks)	500
5–7 Method., data collection, procedures	§5–7	§6: 6.1 FFT background, 6.2 set-up, 6.3 data collection, 6.4 procedures	600
8 Data	§8	§7 raw data (+ resolved? column)	180
10 Processed data	§10	§8 analysis: table, ln–ln graph, Y vs d^{-5} , f_1 vs d , comparison	420
11 Discussion & conclusion	§11	§9 discussion (preliminary)	230
12 Evaluation	§12	§10 evaluation: §10.1–10.5 systematic, §10.6 random	500
–	–	§11 further methodology (INTERNATIONAL §11)	200
–	–	§12 further data & discussion (INTERNATIONAL §12–13)	320
–	–	§13 numerical simulation, RK4 (INTERNATIONAL §14)	250
11.2, 13 Conclusion, extension	§11.2, §13	§14 conclusion & extensions	220
A–E Appendices	A–E	A–H (§9 of this plan)	0

Structure at a glance

Edwin skeleton
 INTERNATIONAL addition
 v2 emphasis

§1 Introduction (motivation paragraph from INTL §1.4)	350
§2 Qualitative prediction	70
§3 Hypothesis <i>quantitative</i> : slope = -5	50
§4 Variables — FFT settings as controlled variables	60
§5 Theoretical model (moved early; + INTL-style remarks §5.4)	500
§6 Methodology — §6.1 FFT background mirrors INTL §6.1	600
§7 Raw data (+ <i>resolved?</i> column, half-range uncertainty)	180
§8 Analysis: processed table, ln–ln graph, Y vs d^{-5} , f_1 vs d	420
§9 Discussion (preliminary) — hand over to evaluation	230
§10 Evaluation: systematic §10.1–10.5, random §10.6 (sign, size, remedy)	500
§11 Further methodology (INTL §11)	200
§12 Further data & discussion (INTL §12–13)	320
§13 Numerical simulation — RK4 of the full equations (INTL §14)	250
§14 Conclusion & extensions	220
Appendices A–H (derivations, propagation, code, extra tables)	0

Figure 2: The essay spine. Colours show where each section comes from; the italic numbers are word budgets.

3 Section-by-section content

3.1 1 Introduction

1.2 Modes of oscillation and beats (≈ 150 words). Same ideas as Edwin’s 1.2 but for translational displacements x_1, x_2 of the magnets: in-phase mode (f_1 : magnets move together, separation constant, magnets “invisible”); anti-phase mode (f_2 : separation oscillates, magnetic force does work). Any release is a superposition, hence beats with $f_{\text{avg}} = (f_1 + f_2)/2$ and $f_{\text{beat}} = (f_2 - f_1)/2$. Include the identity

$$4 f_{\text{avg}} f_{\text{beat}} = f_2^2 - f_1^2, \quad (1)$$

which ties your test quantity to Edwin’s average/beat language and explains why the beats vanish at large d ($f_2 \rightarrow f_1$). Figure 2: your own sketch of the two modes (do not copy the IYPT slides). Figure 3: one of your own $x(t)$ traces showing beats.

1.3 Motivation & RQ (≈ 150 words). Origin: IYPT 2023 problem 6 “Magnetic Pendulums” (cite the problem statement); analogy to contactless/magnetically coupled resonators; personal angle (coupling can be tuned continuously by moving the springs, unlike a fixed spring). State the RQ.

3.2 2 Qualitative prediction & 3 Hypothesis

Repulsion $\propto d^{-4}$, so the stiffness added to the anti-phase mode, $\gamma = -dF/dd \propto d^{-5}$, collapses quickly with d : $f_2 \rightarrow f_1$ and the beat period $\rightarrow \infty$; f_1 stays constant.

Hypothesis: f_1 is independent of d ; $f_2^2 - f_1^2 \propto d^{-5}$, i.e. $\ln(f_2^2 - f_1^2)$ against $\ln d$ is a straight line of gradient -5 .

3.3 4 Variables (table, Edwin style)

Independent: d (centre-to-centre $= g+t$; magnets mounted, measured at equilibrium). **Dependent:** f_1, f_2 (hence $f_2^2 - f_1^2$). **Controlled:** same spring pair; clamped length L ; clamp torque; same magnet pair and orientation; initial displacement (magnitude and which spring); release method; camera fps, resolution and distance; record length T ; FFT settings (window, N , zero-padding) — INTERNATIONAL lists FFT settings as controlled variables, do the same. **Uncontrolled:** air currents; temperature (E of copper); residual clamp slip.

3.4 5 Theoretical model (≈ 500 words)

5.1 Leaf spring as an oscillator. Cantilever tip stiffness $k = 3EI/L^3$; effective mass $m = M + \frac{33}{140}\mu L$ (Rayleigh), so $\omega_0^2 = k/m$ (derivation in Appendix A, or cite). One sentence: the Euler–Bernoulli continuum model of the IYPT slides reduces to this for a heavy tip mass; a finite-element check shows $< 1\%$ difference in $f_2^2 - f_1^2$ for $M/(\mu L) \geq 0.4$ (numbers in §10). **Recommended (v2):** do not rely on E at all — pluck each bare spring and use $f = \frac{1}{2\pi}\sqrt{k/m}$ to fix k/m for that spring directly. The static load–deflection measurement then only cross-checks k , and the temperature dependence of E , the Rayleigh factor and the lumped approximation all drop out of the final theory of f_1 .

5.2 Magnetic force. Coaxial identical dipoles, like poles facing (from $\mathbf{F} = \nabla(\boldsymbol{\mu} \cdot \mathbf{B})$, Appendix B):

$$F(r) = \frac{3\mu_0\mu_m^2}{2\pi r^4} \quad (\text{repulsive}). \quad (2)$$

Equilibrium: springs bend outward by δ with $k\delta = F(d)$. You measure d directly, so the IYPT quintic for the equilibrium point is not needed — say so.

5.3 Linearised coupling and normal modes. $F(d + x_2 - x_1) \approx F(d) - \gamma(x_2 - x_1)$ with

$$\gamma = -F'(d) = \frac{6\mu_0\mu_m^2}{\pi d^5}. \quad (3)$$

Equations of motion (x_1, x_2 positive in the same lab direction, cf. Fig. 4; m = effective oscillator mass, M = magnet mass, μ_m = magnetisation moment, μ = linear density):

$$m\ddot{x}_1 = -kx_1 + \gamma(x_2 - x_1), \quad m\ddot{x}_2 = -kx_2 - \gamma(x_2 - x_1). \quad (4)$$

Normal coordinates $\eta = x_1 + x_2$, $\xi = x_1 - x_2$ (matrix form exactly like Edwin’s eqs. 12–15):

$$\omega_1^2 = \frac{k}{m}, \quad \omega_2^2 = \frac{k + 2\gamma}{m} \implies f_2^2 - f_1^2 = \frac{\gamma}{2\pi^2 m} = \frac{3\mu_0\mu_m^2}{\pi^3 m d^5}. \quad (5)$$

Linearised forms for graphs: (i) $\ln(f_2^2 - f_1^2) = \ln C - 5 \ln d$ with $C = 3\mu_0\mu_m^2/(\pi^3 m)$; (ii) $f_2^2 - f_1^2 = Cd^{-5}$ (straight line through the origin against d^{-5} ; compare its gradient with C from independently measured μ_m, m).

5.4 Remarks (INTERNATIONAL §5.3 style). Dimensional check $[\mu_0\mu_m^2/(md^5)] = \text{s}^{-2}$. List the assumptions you will revisit: (a) point dipoles, (b) linearised force (small amplitude), (c) identical springs, (d) lumped mass, (e) no damping, (f) magnets stay coaxial, (g) the FFT returns the true frequencies.

3.5 6 Methodology (≈ 600 words)

6.1 FFT background — INTERNATIONAL §6.1 adapted to *frequency* measurement; the physics you need is in §4 of this plan.

6.2 Set-up & apparatus: labelled photo (Edwin Fig. 3), apparatus list, magnet specification (grade, $D \times t$, mass), spring dimensions (L, b, h), camera (model, fps, resolution), tracking software.

6.3 Method of data collection — narrate like Edwin §6: Method 1 (read beat and average periods from $x(t)$) failed at large d because the envelope disappears when $f_2 - f_1$ is small; Method 2 (FFT of $x(t)$) adopted. Show one $x(t)$ and its FFT with the two peaks labelled (Edwin Figs. 4–6).

6.4 Procedures:

1. spring stiffness k : static load–deflection, plot F vs x , LINEST (Edwin §7.1);
2. magnet moment μ_m : either datasheet B_r ($\mu_m = B_r V / \mu_0$) or, better, one magnet on an electronic balance, the other brought coaxially above it at measured r , $F = \Delta mg$, fit $F \propto r^{-4}$ (doubles as an independent check of the r^{-4} law);
3. per-spring dynamic calibration: f_a, f_b of each spring alone (pluck, FFT). These give k/m per spring with no E , no 33/140 and no lumping assumption — they are also the two detuned frequencies used in §10.4;
4. masses M, μ ;
5. f_1, f_2 vs d : set d , displace one spring by $A = 10$ mm, release, record T s at f_s fps, track, detrend, FFT, read the two highest peaks, 3–5 trials per d .

3.6 7 Raw data (≈ 180 words)

Table 1 basic quantities ($L, b, h, M, \mu, k \pm \Delta k, \mu_m \pm \Delta \mu_m, m; f_a, f_b$). Table 2 F vs x for k . Tables 3/4 f_1 and f_2 per trial with mean and half-range (justify half-range vs standard deviation as INTERNATIONAL §8.2 does: $n = 3-5$ is too small for SD). State T, f_s, N and the bin width $1/T$ in the caption and add a column “resolved?” ($f_2 - f_1 \geq 2/T$). Rows $d = 40-60$ mm: “peaks unresolved” — this is a *result*, not a gap.

3.7 8 Analysis (≈ 420 words)

8.1 Processed table: $d, \ln d \pm \Delta d/d, Y = f_2^2 - f_1^2, \Delta Y = 2(f_1 \Delta f_1 + f_2 \Delta f_2), \ln Y \pm \Delta Y/Y, d^{-5} \pm 5(\Delta d/d)d^{-5}$. One sample calculation (Edwin style).

8.2 Graph 1: $\ln Y$ vs $\ln d$ with error bars on *both axes* ($\Delta(\ln d) = \Delta d/d \approx 2-5\%$ — see §4.7), best- and worst-fit lines (or LINEST if the bars are too small). Report the gradient -3.75 ± 0.19 (unweighted) and note the weighted value -3.48 ± 0.24 — the disagreement itself exposes the $\ln$ -transform problem.

8.3 Graph 2: Y vs d^{-5} (theory: straight line through the origin, gradient C). Graph 3: f_1 vs d (theory: horizontal) — you will see the 2.6% drift.

8.4 Comparison table (Edwin Tables 13/14): theoretical gradient -5 (exact) vs experimental; theoretical C from μ_m, m vs the experimental gradient of Graph 2; percentage differences.

3.8 9 Discussion — preliminary (≈ 230 words)

Trend confirmed (Y falls steeply, $f_2 \rightarrow f_1$), but: exponent $-3.75 \neq -5$; f_1 not constant; points beyond 35 mm unresolved; residuals systematic (curvature) rather than random. Frame the four candidate causes and hand over to §10.

3.9 10 Evaluation (≈ 500 words)

One sub-section per effect, each with **sign, size, remedy**.

- **10.1 FFT resolution & leakage** (methodological/procedural). Two tones closer than $\sim 1/T$ (rectangular) or $\sim 2/T$ (Hann) merge or “pull” each other; $f_2 - f_1 = 0.039$ Hz at 35 mm needs $T \geq 26-51$ s, and 0.009 Hz at 40 mm needs $T \geq 110-220$ s (table in §4.1).

- **10.2 Amplitude nonlinearity** (the linearisation itself). The anti-phase oscillation modulates the separation, γ is not constant over a cycle, and the oscillation samples a softer-than-dipole force. Perturbation estimate (Appendix D), RK4 (§13) and amplitude-segment analysis of the data (f_2 vs A , extrapolate $A \rightarrow 0$). Include the tilt estimate: the tip rotation is $\theta \simeq 3A/(2L) \approx 6^\circ$ at $A = 10$ mm, $L = 150$ mm, so $\cos \theta \approx 0.5\%$ and the coaxial force is slightly overestimated — one line of estimate, one RK4 check.
- **10.3 Finite magnet size.** The dipole law is the $d \gg$ magnet-size limit; your $d/D \approx 2$ –6. Charged-disc model (Appendix E): the *apparent* $\ln$ – $\ln$ gradient over 21–35 mm falls from -4.9 ($D = 6$ mm) to -3.75 ($D = 20$ mm) — the data are consistent with a soft force law at short range.
- **10.4 Non-identical springs (detuning).** f_1 drifting up by 2.6% as d decreases is exactly what the detuned model predicts: $\omega_{1,2}^2 = \bar{\omega}^2 + \kappa \mp \sqrt{\Delta^2 + \kappa^2}$ with $\kappa = \gamma/m$ — as the coupling grows, both modes are pushed towards $\bar{\omega}$. Fit f_a, f_b (from §6.4.3) with $n = 5$ fixed: reproduces $f_1 : 0.798 \rightarrow 0.819$ Hz.
- **10.5 Damping & other minor effects:** damping linewidth $\approx 1/(\pi\tau)$ (see §4.5); copper creep/temperature; measurement of d ($\Delta d = 0.5$ mm gives 2.4% on Y at 21 mm).
- **10.6 Random errors:** peak-reading scatter; release impulse; d placement; mitigated by 3–5 trials, half-range uncertainty.

3.10 11 Further methodology (≈ 200 words)

INTERNATIONAL’s “further methodology” move, adapted: (1) longer records $T \geq 90$ s; (2) $A = 3$ mm release (plus one $A = 10$ mm run for the amplitude study); (3) track *both* magnets and form the normal coordinates $\eta = x_1 + x_2$ (removes the common-mode motion and the drift), $\xi = x_1 - x_2$ (pure anti-phase); (4) three windows (rect / Hann / Blackman — INTL-style comparison), and a time-domain two-damped-sinusoid least-squares fit (beats the $1/T$ limit; gives a statistical Δf).

3.11 12 Further data & discussion (≈ 320 words)

Table: $\text{gradient} \pm \Delta$ for each method (rect / Hann / Blackman / two-tone fit / normal-coordinate FFT) — like INTERNATIONAL Figs. 26–27. Graph: f_2 vs amplitude (segments) with extrapolation to zero amplitude. Graph: $\ln Y$ vs $\ln d$ for old vs new data with both theory curves (point dipole and finite size). Discuss which correction moved the gradient towards -5 and by how much; be explicit if a residual discrepancy remains and what it implies (e.g. “the dipole law is valid only for $d \gtrsim 4D$ ”).

3.12 13 Numerical simulation (≈ 250 words)

RK4 integration of the *full* equations (optionally with the finite-size F), using your measured k, m, C_F :

$$m\ddot{x}_1 = -kx_1 - [F(d + x_2 - x_1) - F(d)], \quad m\ddot{x}_2 = -kx_2 + [F(d + x_2 - x_1) - F(d)], \quad (6)$$

with $F = C_F r^{-4}$ (signs per Fig. 4 convention). Two uses: (i) predict $f_2(A, d) \rightarrow$ the amplitude correction used in §12; (ii) generate synthetic $x(t)$ at your fps and T and push them through the *same* FFT pipeline $\rightarrow$ isolates the bias caused purely by the analysis (in my test at $T = 40$ s, window peak-picking gave gradients -3.3 to -3.5 while the two-tone fit recovered the true value). Code $\rightarrow$ Appendix G.

3.13 14 Conclusion & extensions (≈ 220 words)

Answer the RQ in three sentences (f_1 constant within $\dots$; $f_2^2 - f_1^2$ falls as d^{-n} with $n = \dots \pm \dots$; the dipole law holds for $d \geq \dots$). Name the dominant systematic error and what fixed it. Extensions:

vary μ_m (magnet grade) at fixed d to test the μ_m^2 dependence; attractive orientation (softening $\rightarrow$ instability at a critical d); three springs; eddy-current damping with a copper plate.

4 FFT error analysis — what to write

INTERNATIONAL needed *amplitudes* (SNR), so its corrections were energy-correction factors and baseline-noise subtraction. You need *frequencies*, so the relevant FFT physics is different — say this explicitly.

1. **Resolution.** $X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i k n / N}$; bin spacing $\Delta f_{\text{bin}} = f_s / N = 1/T$. Two tones are separable only if $|f_2 - f_1| \gtrsim 1/T$ (rectangular) or $\gtrsim 2/T$ (Hann; main lobe twice as wide). From your data:

d/mm	$f_2 - f_1/\text{Hz}$	bins at $T = 20\text{ s}$	T_{min} rect/s	T_{min} Hann/s
21	0.240	4.8	4	8
25	0.141	2.8	7	14
28	0.097	1.9	10	21
31	0.074	1.5	14	27
35	0.039	0.8	26	51
40	0.009	0.2	111	222

2. **Leakage & windows.** Rectangular: narrowest main lobe (best for two close peaks of similar height) but -13 dB side lobes; Hann: 4-bin main lobe, -31 dB ; Blackman: 6-bin, -58 dB . Because both your peaks have similar amplitude and there is no strong interferer, the rectangular (or Hann) window is the right choice — the opposite of the usual “Hann by default” advice, and unlike INTERNATIONAL’s case. Amplitude correction factors are *not* needed because you only read peak positions.
3. **Peak pulling.** When main lobes overlap, the two maxima are pushed apart or merge; in my synthetic test ($T = 20\text{ s}$) the errors were $\pm 2\text{--}7\text{ mHz}$ at $21\text{--}31\text{ mm}$, $\pm 10\text{ mHz}$ at 35 mm and total failure at 40 mm — quote your own numbers from `ee_simulations.py` Part A with your real T and f_s .
4. **Scalloping / bin centring.** The true frequency lies anywhere within $\pm \frac{1}{2}$ bin, i.e. $\pm 1/(2T)$, unless you zero-pad and interpolate (parabola through the three bins around the peak). Zero-padding interpolates the spectrum, it does *not* add resolution — make that distinction.
5. **Damping linewidth.** $e^{-t/\tau} \cos(2\pi f t)$ has $\text{FWHM} \approx 1/(\pi\tau)$; with $\tau \approx 25\text{ s}$ that is 0.013 Hz — comparable to $f_2 - f_1$ at 40 mm , so beyond $\sim 35\text{ mm}$ the modes cannot be separated by any spectral method applied to a single coordinate $\rightarrow$ justification for normal-coordinate tracking and time-domain fitting.
6. **Sampling & pre-processing.** $f_s = \text{fps}$ ($30/60\text{ Hz}$) $\gg 2f$: no aliasing; but video frame timing jitters and frames drop $\rightarrow$ resample to a uniform grid before the FFT (the script reports the jitter). Detrend: subtract the mean (or fit and remove a constant) — the equilibrium offset of the magnets puts a DC peak at 0 Hz (cf. the offset term in eq. (4)), and its leakage contaminates the low-frequency region where your small- d peaks are far away but your large- d peaks are not. Archive the raw videos, not only the Tracker exports.
7. **Propagation into your graph.** $\Delta(\ln Y) = 2(f_1 \Delta f_1 + f_2 \Delta f_2)/Y$ grows from 0.04 to 1.1 across your range for $\Delta f = 5\text{ mHz}$, so error bars must be drawn in $\ln$ -space and the fit weighted; otherwise the noisy large- d points dominate the gradient. Do not forget the horizontal error bars: $\Delta(\ln d) = \Delta d/d$. With $\Delta d = 1\text{ mm}$ this is 4.8% at 21 mm — already comparable to $\Delta(\ln Y)$ there — and it propagates as $5 \Delta d/d$ into any d^{-5} comparison.

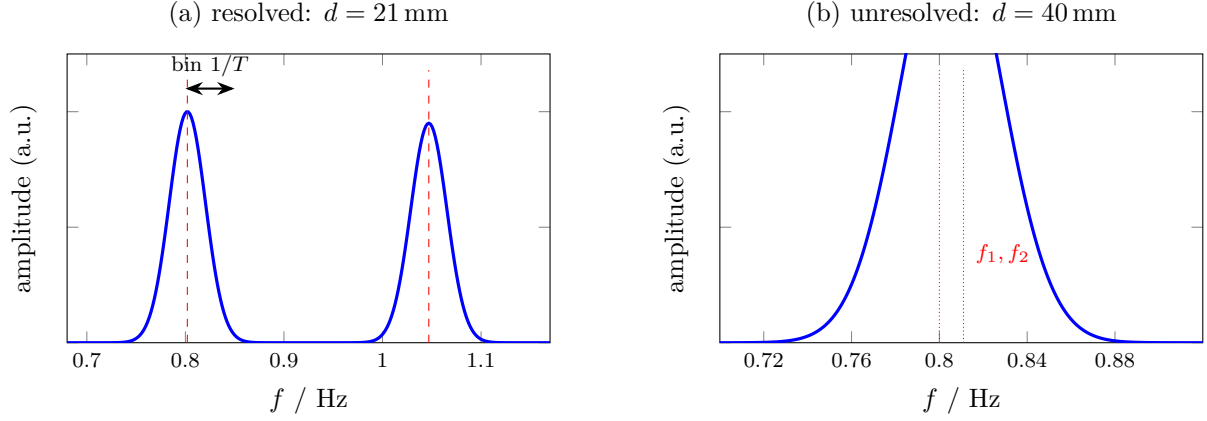


Figure 3: Synthetic two-tone spectra through the same FFT pipeline as the data ($T = 20$ s). Left: separation 0.245 Hz ≈ 5 bins — both peaks read. Right: separation 0.009 Hz = 0.2 bins — the two true frequencies (dotted) are unresolvable; only one merged peak exists. This is the mechanism behind the “#NUM!” rows at $d \geq 40$ mm.

5 “Non-linearisation” improvements — three levels (use all three, in order)

Level 1 — analysis (cheap, do it now).

Replace “take logs then LINEST” with a weighted non-linear least-squares fit of $Y = C d^{-n}$ (and $Y = C d^{-5}$ with n fixed) to the *un-transformed* data. Advantages to state: no point is discarded because $\ln(0)$ or $\ln(\text{negative})$ fails; heteroscedastic errors are handled by the weights; C and n come with proper uncertainties. With your 6 points: $n = 3.5 \pm 0.2$ (vs $\ln\text{-}\ln -3.75$) — report both and explain the difference.

Level 2 — physics (the systematic-error story).

The model linearised $F(r)$ about d . Keep the full r^{-4} (or finite-size) force: perturbation formula (Appendix D) in the body, RK4 (§13) for exact numbers, amplitude-segment analysis + extrapolation to $A \rightarrow 0$ for the data. Tabulate the size of the correction at each d .

Level 3 — model (finite size + detuning).

Fit models (c) and (d) of §11 step 5: (c) finite-size charged-disc force; (d) detuned springs. Present the comparison of residual patterns / reduced χ^2 as the decisive evidence for which assumption failed. Report reduced χ^2 (or simply χ^2/dof) for each fit, not only R^2 .

6 Lab / data checklist (you have lab access and the raw tracks)

- ☐ Magnet dimensions D, t , mass M , grade (B_r) — needed for §10.3 and the theoretical C .
- ☐ Spring L, b, h, μ (weigh a known length), k by static loading; f_a, f_b of each spring alone — the primary calibration (§6.4.3).
- ☐ Balance measurement of $F(r)$ for 4–6 values of r (checks r^{-4} and gives μ_m).
- ☐ Re-run $d = 21 \dots 60$ mm with both magnets tracked, $T \geq 90$ s, $A = 3$ mm (and one $A = 10$ mm run each). Name files `d21_trial1.csv` etc.
- ☐ `python analyse_tracks.py "tracks/*.csv" -out results` → per-file diagnostic figure ($x(t)$ + two-tone fit, three-window FFT, f_2 vs amplitude), `summary.csv`, and the $\ln\text{-}\ln$ gradient per method. Then form η, ξ from the two tracked magnets and run the same script on them.
- ☐ Archive raw videos per run (examiner authenticity).
- ☐ Sanity check of magnitudes: $\gamma = 2\pi^2 m Y$ from your data vs $\gamma = 6\mu_0\mu_m^2/(\pi d^5)$ from μ_m . If they differ by more than a factor ~ 2 , revisit the coaxial-dipole assumption or the definition of d before

writing §8.4.

- Re-derive the $f_2 - f_1$ requirement with your real T : if your original records were $\sim 20\text{--}40$ s, say in §7 that 35 mm was marginal and 40 mm unresolvable.

Suggested writing order (new in v2)

Collect data → write §5 (model) and §6 (methodology) → §4 variables → §7–8 (raw data, analysis, graphs) → §9–13 (discussion, evaluation, further work, simulation) → §1–3 and §14 *last* (abstract/contents last of all). The middle of the essay is stable once the data exist; the ends depend on everything else.

7 Figures and tables (final list)

Figures: 1 set-up sketch (done) · 2 mode sketch · 3 set-up photo · 4 $x(t)$ with beats (small d) · 5 $x(t)$ with no visible envelope (large d) · 6 FFT with two peaks · 7 F vs $x(k)$ · 8 F vs r on the balance (μ_m) · 9 $\ln Y$ vs $\ln d$ (Graph 1) · 10 Y vs d^{-5} (Graph 2) · 11 f_1 vs d (Graph 3) · 12 synthetic-spectrum resolution figure · 13 f_2 vs amplitude · 14 gradient by method (further data) · 15 old vs new data with dipole and finite-size theory · 16 RK4 vs linearised prediction.

Tables: 1 constants · 2 k data · 3/4 f_1, f_2 raw · 5 processed · 6 theory vs experiment (gradient, C) · 7 systematic-error summary (effect, sign, size, remedy) · 8 further raw data (appendix) · 9 gradient/ C per method and per model · 10 simulated vs measured f_2 shift.

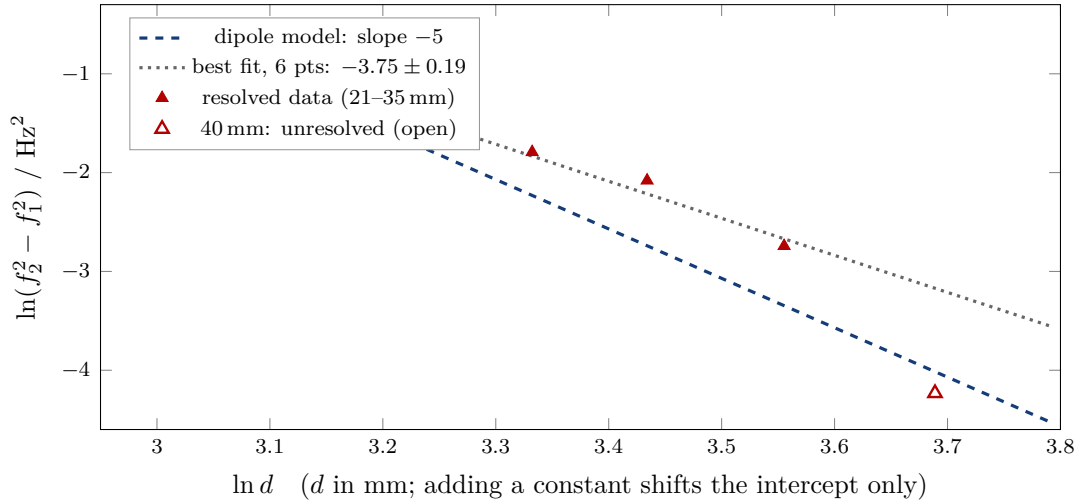


Figure 4: Your current data (triangles) against the dipole prediction and the 6-point best fit. The 40 mm point (open) sits far below the resolved trend — consistent with the resolution limit of Fig. 3b, not with a new physical regime. Note the possible *downward curvature* of the resolved points at small d — the finite-size signature (§10.3).

8 Assessment-criteria map (new in v2)

Criterion	Where this plan earns it
A: Focus & method	RQ (§1); variables incl. FFT settings (§4); justified method choices, attempt 1 → attempt 2 narrative (§6.3).
B: Knowledge & understanding	Full derivation chain (§5); FFT/DFT theory (§6.1, §4 of this plan); correct physics vocabulary throughout.
C: Critical thinking (12 marks)	Quantitative hypothesis; comparison with theory (§8.4); error propagation & weighting (§8.1–8.2); <i>two distinct error types</i> (measurement §10.1 vs model §10.2–10.5) each with sign, size, remedy; improvements quantified (§12); simulation verification (§13).
D: Presentation	Numbered figures/tables (§7 of this plan); captions that state the point of each figure; derivations and bulk data pushed to appendices A–H.
E: Engagement	Personal angle and IYPT origin (§1.3); honest reflection on residual discrepancy (§12); independence evidenced by your own scripts (§6 of this plan).

9 Appendices to write

- A** Cantilever stiffness $3EI/L^3$, effective mass $\frac{33}{140}\mu L$, and FEM check of lumped vs beam model ($< 1\%$). + *dynamic calibration method* ($f = \frac{1}{2\pi}\sqrt{k/m}$).
- B** Dipole field → force between coaxial dipoles $F = 3\mu_0\mu_m^2/(2\pi r^4) \rightarrow \gamma = 6\mu_0\mu_m^2/(\pi d^5)$.
- C** Detuned normal modes $\omega_{1,2}^2 = \bar{\omega}^2 + \kappa \mp \sqrt{\Delta^2 + \kappa^2}$ and the d^{-5} limit.
- D** Series expansion of $F(d - \xi)$, α and β , Landau–Lifshitz amplitude shift $(a/d)^2(15g/8 - 125g^2/48)$, comparison with RK4.
- E** Finite-size (charged-disc) magnet force and apparent-gradient table.
- F** Uncertainty propagation (ΔY , $\Delta \ln Y$, Δd^{-5} , LINEST worst lines). + *the x -error* $\Delta(\ln d) = \Delta d/d$.
- G** Python: `analyse_tracks.py` (FFT windows, two-tone fit, segments), `ee_simulations.py` (RK4, synthetic FFT), `finite_size_magnets.py`.
- H** Further raw data tables (all windows/methods), as INTERNATIONAL Appendix A.

10 Ready-made numbers from my checks

Replace with your own values once you have real inputs. Assumed in the scripts: f_s , τ , magnet size and mass — marked # ASSUMED.

- $\ln$ - $\ln$ fit, 6 points: gradient -3.75 ± 0.19 , intercept 10.68 (matches your chart); weighted ($\Delta f = 5$ mHz): -3.48 ± 0.24 . Free power law on raw Y : $n = 3.5 \pm 0.2$.
- Non-linear force, $A = 10$ mm release: $f_2 + 9.8\%$ (21 mm), $+1.1\%$ (35 mm); gradient $-5.00 \rightarrow -5.16$. $A = 2$ mm: $+0.3\%/+0.03\%$.
- Finite-size discs, apparent gradient over 21–35 mm: $D = 6$ mm -4.87 ; 10 mm -4.6 ; 12 mm -4.5 ; 15 mm -4.2 ; 20 mm -3.75 ; 25 mm -3.4 ($t = 3$ –6 mm changes these by < 0.1).
- Lumped vs beam: Y differs by 0.03% ($M/\mu L = 3.3$), 0.4% (0.8), 1% (0.4).
- Detuned model fitted with $n = 5$: $f_a = 0.793$ Hz, $f_b = 0.849$ Hz reproduces the f_1 drift $0.798 \rightarrow 0.819$ Hz.
- Synthetic tracks at $T = 40$ s: window peak-picking gradients -3.3 to -3.5 ; two-tone fit -4.9 (true -4.9) → the analysis method alone can shift the gradient by more than 1.

11 Pitfalls to avoid (examiner’s eye)

- Do not present the 40 mm point as data; present the unresolved range as a measured limit of the method.

- Do not claim “ f_1 constant” when it drifts 2.6%; explain it (§10.4).
- Do not copy IYPT slide figures or equations without derivation; cite the IYPT problem and derive the lumped model yourself.
- Keep all corrections quantitative (sign and size) — hand-waving loses evaluation marks.
- Every graph: error bars on both axes, best/worst lines or LINEST uncertainty, units, gradient with uncertainty, theory line overlaid where possible.
- Word count: equations, tables, captions, appendices excluded — move derivations there ruthlessly.

v2 — green text marks additions/changes over the previous plan. Regenerate every placeholder number with your own inputs before it enters the essay.